The Hong Kong Polytechnic University
School of Professional Education and Executive Development

SEHS3245 Web Systems and Technologies
Semester 1, 2024-2025

Project and Presentation

Group Project (30% of the assessment of this subject)

This group project aims:

- To conduct **research** on relevant reference materials or sources.
- To design web pages and web server programs using relevant web systems and technologies.
- To manage your group assignment efficiently and effectively in **teamwork** within the given schedule.

Project Instructions:

1) Students have to form their own group with **4 to 6 members**, who come from the **same lecture class 101, 102, 103 or A01.** Members in a group can come from **different tutorial groups** of the same lecture class.
2) Choose one member in your group to be the group leader.
3) Fill in the **group registration form**.
4) The **group leader** uploads the **completed group registration form** to Blackboard by **10PM, 28 September 2024 (Saturday)**.
5) If your name is not in the group registration form, the lecturer will form a group for you and/choose the group leader for your group.
6) All members in a group should exchange their contact information such as phone number, email address, … with each other for communication and collaboration.

Project Specification:

- Choose an online shop selling products (e.g., books, clothes, electrical appliances, electronics, Internet of things, …) or services (healthcare services, financial technology services, data center services, cloud computing services, …) or both.
- Develop 6 web pages as follows:
  - The first web page is an "introduction" web page which is designed to introduce your online shop. The number of words for introduction should **NOT** exceed 600 words. This first web page contains a logo of your online shop created by you and hyperlinks/buttons to connect to the second, third or fourth web page.
  - These second, third and fourth web pages are "shopping" web pages which describe the products or services or both. Each of these web pages contain **FIVE** products/service

plans including their unit prices, quantities, …. A user can navigate through any of these "shopping" web pages and even go back to the first "introduction" web page. The user can choose any of the products/service plans and click a "shopping cart" button at any of these web pages (i.e., the second, third and fourth web pages) which links to the fifth web page.

- The fifth web page is a "shopping cart" web page which contains the products/service plans chosen by the user. This web page allows the user to go back to the "shopping" web pages to change the chosen products/service plans. This "shopping cart" web page also contains a "buy" button which is supposed to link to a payment gateway. In this project, there is no need to implement the payment process through a payment gateway. Once the user presses the "buy" button, the sixth web page appears.
- The sixth web page just shows "thank you" message designed by you and stores the products/service plans chosen in a MySQL table.

Project Implementation:

- Install Apache web server.
- Use HTML5 following XHTML syntax rules, CSS3, JavaScript to develop client-side web pages.
- Use PHP to develop web server programs.
- Use MySQL for manipulating data.
- Test your web pages and programs with Apache and MySQL.
- Throughout the web page development, no cookies are allowed. Session tracking should be used instead.

Project Submission:

1. Put the source codes (e.g., codes at .html, .css, .js and .php files) and the MySQL statement(s) used (e.g., Create Table statement) into a Microsoft Word 2016 document. The Microsoft Word 2016 document should also contain this subject code (i.e., SEHS3245), this subject name (i.e., Web Systems and Technologies), submission date, all members' full names and student ID numbers on the top of the first page.
2. Put the .html, .css, .js, .php files and the Microsoft Word 2016 document into a folder.
3. Make a zip file of the folder containing all your work:
   - Click on the folder and go ***outside*** of the folder where your work is saved
   - Right-click on the folder that contains all your work, then, depending on your computer, do EITHER of the following:
     - Select **7-Zip** then select the second-last option **Add to "project.zip"** (assuming that you are in a folder called **project**)
     - Select **Send-to** and then **Compressed (zipped) folder**
   - A new **.zip** file should be created
   - Note: **7-zip** is a freeware downloadable through Internet.

4. Right-click on this **.zip** file, then select **Rename**, then change the name to be the group number (to be provided by the lecturer) and the group leader name. If the group number is 3 and the group leader name is Chan Tai Man, then the filename should be **3_ChanTaiMan.zip**
5. The **group leader** submits the **.zip** file to Blackboard by **10PM, 30 November 2024 (Saturday)**.
6. Note: no videos and no audios are allowed in this project submission.

While submitting the softcopy via Blackboard, a timestamp will be placed on the softcopy of your assignment. There is a sharp cut-off time at Blackboard, so late submission will be recorded at Blackboard. Softcopy submitted via email or other means will NOT be accepted unless Blackboard is not available. As many students will submit their assignments to Blackboard at around the deadline time, it normally takes longer for uploading your assignment, so it is strongly suggested that you start submitting earlier, say at least 45 minutes before the deadline. Marks will be deducted for late submission. Successful submission of this assignment includes the submission of all the items specified in this section.

Peer Evaluation:

Upon completion of the assignment, **each** student will have to fill in a peer evaluation form (to be downloaded from Blackboard) on self-evaluation and performance of the members in the same group and send the completed form to the lecturer via the Blackboard's Assignment Submission System. In case the Blackboard's Assignment Submission System is not available, send the completed peer evaluation form to the lecturer via e-mail. 25% of marks will be deducted for NOT submitting the completed peer evaluation form.

Peer Evaluation Submission:

- **Each student** fills in the **peer evaluation form**.
- **Each student** uploads the **completed peer evaluation form** to Blackboard by **10PM, 30 November 2024 (Saturday)**.

Project Marking Scheme:

Correctness • Accurate functionality, including validation and realistic details • HTML5, CSS3, PHP, JavaScript and SQL used properly • Specification (above) is followed	50 marks
Readability * • See details below	20 marks
Design	30 marks

• Website looks good and neat (more marks for professional looking website with intuitive and user-friendly interface)	
Total	**100 marks**

*** Readability**

- Commenting is expected for JavaScript code as well as complex or potentially ambiguous code in HTML5.
- Marks are awarded for using good coding style, including, but not limited to:
  - Following standard coding conventions, including XHTML and HTML5
  - Proper indentation
  - Simplicity and clarity
  - Avoiding code repetition
- Marks will be deducted for unnecessary code:
  - Writing extra code that makes the code easier to understand is considered unnecessary.
- Extremely bad coding style or grossly unnecessary complexity may also result in loss of marks.
- If you copy code or comments from your notes or lectures, and do not meet the specification, marks may also be deducted.
- Full marks will only be awarded for good coding style with no ambiguity.

The content of the web page will be marked. The design of the web page using the knowledge you have learned in this subject will also be marked, Use of graphics, background, table and highlighting effect will be marked but **DO NOT** include any video or audio effect to the web pages. Use of hyperlinks to provide a convenient way to navigate around different sections in the web page will be marked.

Plagiarism will be penalized severely. Marks will be deducted for assignments that are plagiarized in whole or in part, regardless of the sources.

Mark deduction items include spelling mistakes, poor writing style, copy and paste directly from the web resources or any other resources.

Late submission is liable to a **penalty of 5%** of the available marks for **each day late**; Saturdays, Sundays and holidays are counted.

Individual Presentation (15% of the assessment of this subject)

This individual presentation aims:

- To demonstrate effectively and attractively your web page and web server program design.
- To explain clearly and concisely the program logic of your web pages and web server programs.

Presentation Instructions:

1) Each student makes a 2-minute (about 5-slide) Microsoft PowerPoint 2016 file that covers the following:
   - Introduce yourself (just your full name).
   - Demonstrate the important operations and features developed by you.
   - Explain the important program logic developed by you.

   The student also presents this orally in English in class.

Presentation Submission:

- Name the Microsoft PowerPoint file to be your group number appended with your name and your student ID number. If your group number is **3**, your name is **Chan Tai Man** and your student ID number **24012345S**, then the filename is **3_ChanTaiMan_24012345S.pptx**.
- **Each student** submits the **.pptx** file to Blackboard by **10PM, 30 November 2024 (Saturday)**.

Presentation Marking Scheme:

Demonstration • Content, Presentation Skills, Time Management	50 marks
Explanation • Clarity and Logic	50 marks
Total	**100 marks**

Late submission is liable to a **penalty of 5%** of the available marks for **each day late**; Saturdays, Sundays and holidays are counted.

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